

Deploying Video + Text Search Systems

Bridging Natural Language Queries with Visual Content



Why Video + Text Search?

Rich but Complex Modality

Video content contains vast information but remains difficult to index effectively

Natural Language Interface

Users expect to **search video with intuitive text prompts** similar to how they search the web

Cross-Domain Applications

Media archives, security systems, educational platforms, and e-commerce all benefit from multimodal search

Technical Requirements

Building effective systems requires **multimodal embeddings + scalable infrastructure** working in tandem

Core Pipeline Architecture



Ingest Video

Segment into meaningful shots, clips, and representative frames



Preprocess

Extract keyframes + generate transcripts with ASR



Embedding

CLIP for visual frames, BERT/LLM for text transcripts



Indexing

Vector DB for efficient multimodal storage and retrieval



Query

Encode text prompt → retrieve matching video segments

Infrastructure Demands



Storage

Petabyte-scale video storage with comprehensive metadata tracking

Compute

GPU-accelerated processing for frame analysis and transcript embedding

Indexing

[FAISS](#)/[Pinecone](#)/[Weaviate](#) for efficient vector similarity search

Serving

FastAPI/gRPC endpoints with robust caching layers for consistent performance

Latency & Scaling Challenges

1-2s

Query Latency
Target

Multimodal searches have
higher computational
demands than text-only

100TB+

Storage Bandwidth

Raw video retrieval strains
network resources during
peak usage

24/7

Stream Processing

Real-time indexing for live
streams requires continuous
compute allocation

99.9%

Availability Target

Multi-tenant systems need
robust resource isolation to
prevent cross-tenant
impacts

Optimizations for Scale & Performance

1 Precompute Embeddings

Process static archives in batch to minimize on-demand computation

2 Hierarchical Indexing

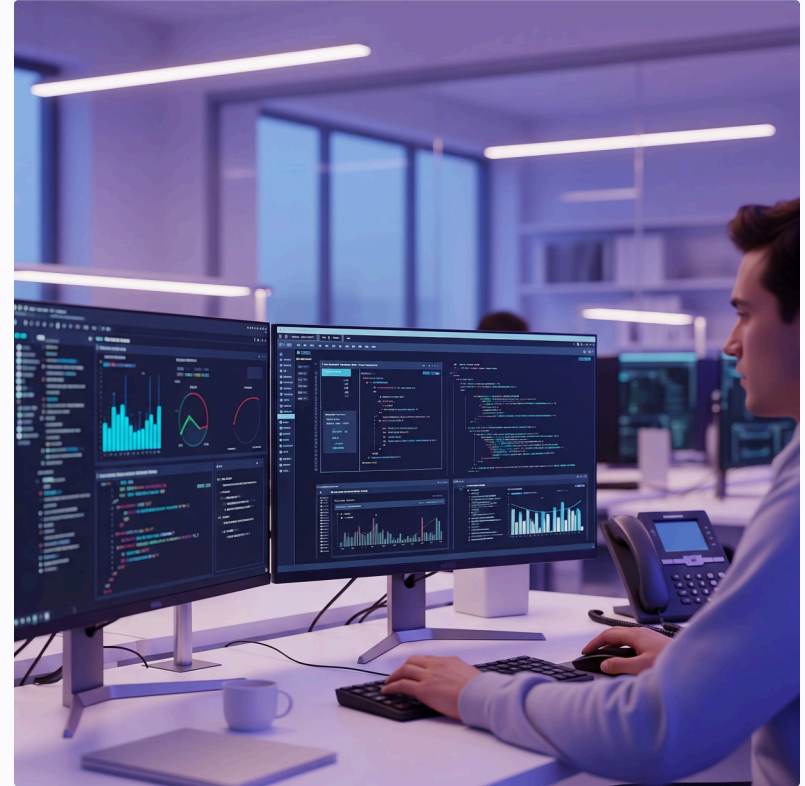
Filter by metadata → query embeddings → apply re-ranking

3 Vector Compression

Compress/quantize embeddings to reduce memory footprint by 4-8x

4 Result Caching

Store frequent queries and results in Redis for sub-50ms responses



Real-World Use Cases



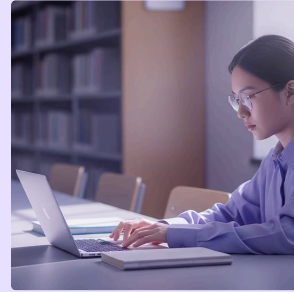
Media

"Find all clips with cats and people laughing" for content creation and editing



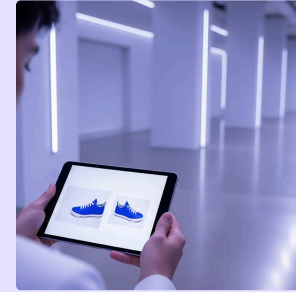
Security

Search CCTV archives for "red car near entrance" to identify suspects



Education

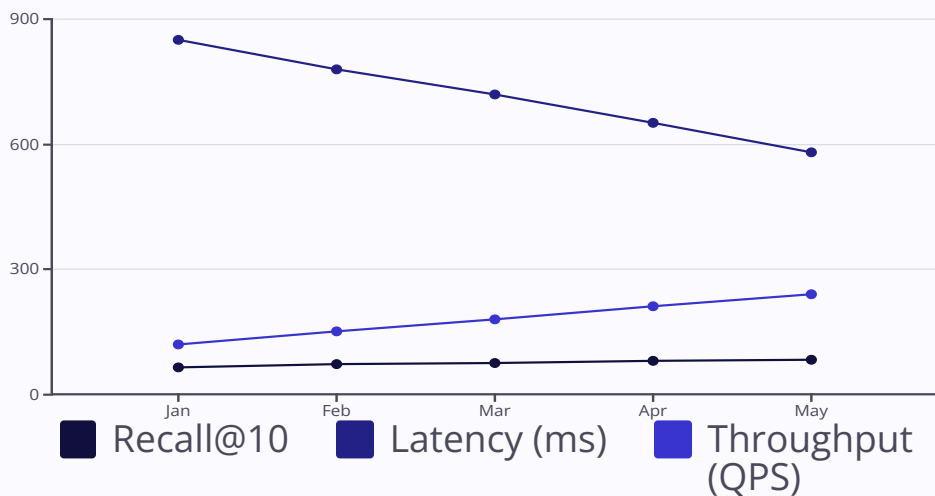
Retrieve lectures mentioning "quantum computing" for focused study



E-commerce

Video search for "blue sneakers on runway" to find trending products

Monitoring & Evaluation



Critical Metrics

- **Recall@k:** Percentage of relevant results returned
- **Query latency:** End-to-end response time
- **System throughput:** Queries per second handled
- **User engagement:** Click-through rates for search quality

Cross-Modal Evaluation

Separately benchmark performance across **text-only, video-only, and cross-modal queries** to identify optimization opportunities

Best Practices for Robust Systems



Dual Representation

Store both **frame embeddings** + **transcripts** to enable multi-pathway retrieval



Hybrid Retrieval

Combine keyword + semantic search for robustness across query types



Intelligent Partitioning

Segment indexes by domain or time period for improved scalability



Continuous Improvement

Regularly retrain with user feedback & incorporate new content types

- ❏ **Engineering Tip:** Maintain separate embedding models for different content domains (e.g., sports vs. education) to improve relevance and reduce semantic drift.

Key Takeaways

1 Unified Architecture

Video+Text search requires **multimodal embeddings** + robust retrieval infrastructure

2 Core Challenges

Address **storage, compute, latency, and scale** simultaneously

3 Optimization Strategy

Combine precomputing, compression, hybrid search, and caching for best results

4 High-Value Applications

Enables transformative AI capabilities for media, security, education, and commerce



Next Steps: Schedule technical architecture review and benchmark against sample video corpus